Freyd Mitchell Embedding Theorem

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The goal is to understand the way of embedding any small Abelian Category into the category of R-modules over some (not necessarily commutative) ring R.

1 Abelian Category

1.1 Basid Structure of Abelian Category

There are multiple ways of defining Abelian Category, here chose the definition used in the book.

First, let $\mathcal{A}$ be a category with a zero object (an object that's both initial and final), denoted as $\mathbf{0}$. We first put on the notion of kernel and cokernel:

Definition 1. *Given $A, B \in \mathbb{A}$ two objects, a* ***zero morphism from** A **to** B* *is the unique morphism obtained by the composition $0 : A \to \mathbf{0} \to B$.*

Let $\phi : A \to B$ be a morphism in $\mathcal{A}$, a morphism $\ker(\phi) : K \to A$ is a ***kernel of** ϕ*, if $\phi \circ \ker(\phi) = 0$, and for every morphism $f : K' \to A$ satisfying $\phi \circ f = 0$, there exists a unique morphism $f' : K' \to K$, such that $f = \ker(\phi) \circ f'$. Namely, the following diagram commutes:*

$$\begin{tikzcd}
K \arrow[r, "\ker(\phi)"] & A \arrow[r, "\phi"] & B \\
K' \arrow[u, dashed, "\exists! f'"] \arrow[ur, "f"'] \arrow[urr, bend right, "0"'] & &
\end{tikzcd}$$

Similarly, a morphism $\operatorname{coker}(\phi) : B \to C$ is a ***cokernel of** ϕ*, if $\operatorname{coker}(\phi) \circ \phi = 0$, and for every morphism $g : B \to C'$ satisfying $g \circ \phi = 0$, there exists a unique morphism $g' : C \to C'$, such that $g = g' \circ \operatorname{coker}(\phi)$. Namely, the following diagram commutes:*

$$\begin{tikzcd}
A \arrow[r, "\phi"] \arrow[drr, bend right, "0"'] & B \arrow[r, "\operatorname{coker}(\phi)"] \arrow[dr, "g"'] & C \arrow[d, dashed, "\exists! g'"] \\
& & C'
\end{tikzcd}$$

Which, the composition of any morphism with the zero morphism is a zero morphism (as the morphism through $\mathbf{0}$ from/to any object must be unique). Also, the universality above guarantees kernels and cokernesl to be isomorphic. Notice that kernels and cokernels under this definitions, are monomorphisms and epimorphisms respectively:

Proposition 1. *All kernels are monomorphisms, and all cokernels are epimorphisms.*

Proof. Let $\phi : A \to B$ be a morphism, $k : K \to A$ denotes its kernel, and $c : B \to C$ denotes its cokernel.

First, for kernel, choose two morphisms $f_0, f_1 : K' \to K$ such that $h = k \circ f_0 = k \circ f_1 : K' \to A$. Then, since $\phi \circ h = (\phi \circ k) \circ f_0 = 0 \circ f_0 = 0$, the universality guarantees a unique morphism $f' : K' \to K$, such that $h = \ker(\phi) \circ f'$. Since f_0, f_1 in the position in f' satisfies the given equality, the uniqueness of f' guarantees $f_0 = f_1 = f'$.

For cokernel, given $g_0, g_1 : C \to C'$ such that $\ell = g_0 \circ c = g_1 \circ c : B \to C'$, as $\ell \circ \phi = g_0 \circ (c \circ \phi) = g_0 \circ 0 = 0$, similar argument on universality guarantees $g_0 = g_1$. $\square$

With this property, kernels and cokernels can be viewed as subobject / quotient objects of certain morphisms.

Now, here is the definition of Abelian Category:

Definition 2. *A category $\mathcal{A}$ is an* ***Abelian Category****, if it satisfies the following axioms:*

- *A0. $\mathcal{A}$ has a zero object $\mathbf{0}$, which serves as both initial and final object.*
- *A1. Finite product and coproduct exists. (also called "direct product" and "direct sum").*
- *A2. Every morphism has a kernel and a cokernel.*
- *A3. Every monomorphism is a kernel of some morphism, and every epimorphism is a cokernel of a morphism.*

With this definition, we can immediately conclude the following:

Proposition 2. *Let $\phi : A \to B$ be a morphism. Then:*

1. *If ϕ is a monomorphism, then $\mathbf{0} \to A$ is its kernel.*
2. *Similarly, if ϕ is an epimorphism, then $B \to \mathbf{0}$ is its cokernel.*

Proof.

1. It's clear that $\mathbf{0} \to A \xrightarrow{\phi} B$ is zero morphism. Now, suppose $f : K' \to A$ satisfies $\phi \circ f = 0 : K' \to B$, then since $0 : K \to A$ satisfies $\phi \circ 0 = 0 : K' \to A$ also, the monomorphism property guarantees $f = 0$. Hence, we have $K \xrightarrow{f} A = K \to \mathbf{0} \to A$ the zero morphism, showing f uniquely factors through $\mathbf{0} \to Ax$ (as the morphism $K \to \mathbf{0}$ is unique).

2. Again, it's clear that $A \xrightarrow{\phi} B \to \mathbf{0}$ is zero. Now, suppose $g : B \to C'$ satisfies $g \circ \phi = 0 : A \to C'$, again $0 : B \to C'$ satisfies $0 \circ \phi = 0 : A \to C'$, showing $g = 0$ using the epimorphism property. So, $B \xrightarrow{g} C' = B \to \mathbf{0} \to C'$ is the zero morphism, showing g uniquely factors through $B \to \mathbf{0}$.

$\square$

Of course, for the most generality, we also have the following properties for any two objects $A, B \in \mathcal{A}$:

- Finite direct sum and direct product coincides, so $A \oplus B$ is a biproduct. Let ι, π denote inclusion / projection respectively, then we require $\pi_j \circ \iota_j = 1_j$ for $j = A, B$, and π_i, ι_j be cokernel / kernel (resp.) for each other if $i \neq j$.

- The homset $\mathrm{Hom}_{\mathcal{A}}(A, B)$ has a natural abelian group structure, with $0 : A \to B$ serves as the identity.

In some lieteerature (for instance, Aluffi), the second property is assumed, and can be used to prove the first property (in fact, the first property is true for general additive category).

Conversely, some literature assumes the first property, and use it to show the second property (for instance, the book of the reference). Here, we'll assume this, and show the abelian group structure on the homsets.

Notation: We'll use $\iota_A : A \to A \oplus B$, $\iota_B : B \to A \oplus B$ to denote the canonical inclusion for coproduct, and $\pi_A : A \oplus B \to A$, $\pi_B : A \oplus B \to B$ to denote the canonical projection for product.

Also, given two morphisms $f_A : C \to A$, $f_B : C \to B$, denote the product morphism as $\begin{pmatrix} f_A \\ f_B \end{pmatrix} : C \to A \oplus B$ (so $\pi_A \circ \begin{pmatrix} f_A \\ f_B \end{pmatrix} = f_A$, and $\pi_B \circ \begin{pmatrix} f_1 \\ f_2 \end{pmatrix} = f_B$). Similarly, given two morphisms $g_A : A \to D$, $g_B : B \to D$, denote the coproduct morphism as $(g_A \ g_B) : A \oplus B \to D$ (so $(g_A \ g_B) \circ \iota_A = g_A$, and $(g_A \ g_B) \circ \iota_B = g_B$).

This is similar to the matrix notation. In general, given objects $A, B, C, D \in \mathcal{A}$, with morphisms $f : A \to C$, $g : A \to D$, $h : B \to C$, and $\ell : B \to D$, we have morphisms $\begin{pmatrix} f \\ g \end{pmatrix} : A \to C \oplus D$ and $\begin{pmatrix} h \\ \ell \end{pmatrix} : B \to C \oplus D$. Which, they gathered into a morphism $s = \left(\begin{pmatrix} f \\ g \end{pmatrix} \ \begin{pmatrix} h \\ \ell \end{pmatrix} \right) : A \oplus B \to C \oplus D$.

Similarly, we can define morphisms $(f \ h) : A \oplus B \to C$ and $(g \ \ell) : A \oplus B \to D$, which further gathered into a morhpism $s' = \begin{pmatrix} (f \ h) \\ (g \ \ell) \end{pmatrix} : A \oplus B \to C \oplus D$.

Notice that $s = s'$, because we have the following four equalities:

$$\pi_C \circ (s \circ \iota_A) = \pi_C \circ \begin{pmatrix} f \\ g \end{pmatrix} = f = (f \ h) \circ \iota_A = (\pi_C \circ s') \circ \iota_A \tag{1}$$

$$\pi_D \circ (s \circ \iota_A) = \pi_D \circ \begin{pmatrix} f \\ g \end{pmatrix} = g = (g \ \ell) \circ \iota_A = (\pi_D \circ s') \circ \iota_A \tag{2}$$

$$\pi_C \circ (s \circ \iota_B) = \pi_C \circ \begin{pmatrix} h \\ \ell \end{pmatrix} = h = (f \ h) \circ \iota_B = (\pi_C \circ s') \circ \iota_B \tag{3}$$

$$\pi_D \circ (s \circ \iota_B) = \pi_D \circ \begin{pmatrix} h \\ \ell \end{pmatrix} = \ell = (g \ \ell) \circ \iota_B = (\pi_D \circ s') \circ \iota_B \tag{4}$$

Which, (1) and (2) state that $s \circ \iota_A = s' \circ \iota_A$ (as their projection onto C and D agrees, so universality of product forces them to be the same); similarly, (3) and (4) state that $s \circ \iota_B = s' \circ \iota_B$ using product property again.

Finally, the two formula regarding the inclusions ι_A, ι_B precisely force $s = s'$ using the coproduct property. So, there's no ambiguity about the order. Which, we'll denote $s = s' = \begin{pmatrix} f & h \\ g & \ell \end{pmatrix}$.

Now, we'll define the "addition" on homsets:

Definition 3. *Let $f, g \in \mathrm{Hom}_{\mathcal{A}}(A, B)$, we define the morphism $f + g \in \mathrm{Hom}_{\mathcal{A}}(A, B)$ to be the following:*

$$A \xrightarrow{\binom{1_A}{1_A}} A \oplus A \xrightarrow{(f\ \ g)} B$$

Another definition is the following:

$$A \xrightarrow{\binom{f}{g}} B \oplus B \xrightarrow{(1_B\ \ 1_B)} B$$

Notice the two actually define the same morphism:

Proof. Consider the morphism $m = \begin{pmatrix} f & 0 \\ 0 & g \end{pmatrix} : A \oplus A \to B \oplus B$, then we have the following:

$$(1_B\ 1_B) \circ m = \tag{5}$$

$\square$

Include: Additive structure on homsets, characterization of direct sums, definition of image / coimage, exactness (and some characterization about image=kernel, coimage=cokernel, and other compositions), subobject, intersection and union.

The goal for this section is establishing some utilities needed for embedding.

Let's first define additive functor:

Definition 4. *Let $\mathcal{A}, \mathcal{B}$ be two Abelian Categories. Let $F : \mathcal{A} \to \mathcal{B}$ be a functor.*
*F is **additive**, if for any two objects $A, B \in \mathcal{A}$, the map $\mathrm{Hom}_{\mathcal{A}}(A, B) \to \mathrm{Hom}_{\mathcal{B}}(F(A), F(B))$ is a group homomorphism with respect to the group structure on each homset.*

For instance, $\mathrm{Hom}_{\mathcal{A}}(A, \_) : \mathcal{A} \to \mathsf{Ab}$ and $\mathrm{Hom}_{\mathcal{A}}(\_, A) : \mathcal{A}^{\mathrm{op}} \to \mathsf{Ab}$ are additive functors, as composition of morphisms respect the additive structure.

Here is a characterization of additive functors:

Theorem 1. *A functor between abelian categories is additive $\iff$ it maps direct sums to direct sums, i.e. $F(A \oplus B) \cong F(A) \oplus F(B)$, with the agreement on inclusions / projections.*

Proof. This requries the characterization of dierct sum: Given $A \oplus B$, then $\pi_j \circ \iota_j = 1_j$ on each $j = A, B$, while $\iota_A \circ \pi_A + \iota_B \circ \pi_B = 1$ on $A \oplus B$.

$\Longrightarrow$: Suppose F is additive. Given $F\pi_j \circ F\iota_j = F(\pi_j \circ \iota_j) = 1_{F(j)}$ for $j = A, B$, and $1 = F(1) = F(\iota_A \circ \pi_A + \iota_B \circ \pi_B) = F\iota_A \circ F\pi_A + F\iota_B \circ F\pi_B$, showing $F(A \oplus B)$ with $F\iota_A, F\iota_B, F\pi_A, F\pi_B$ form a direct sum system for $F(A), F(B)$, hence $F(A \oplus B) \cong F(A) \oplus F(B)$.

$\Longleftarrow$: Suppose F maps direct sum to direct sum, then for every $f, g \in \mathrm{Hom}_{\mathcal{A}}(A, B)$, we have $f + g : A \to B$ being the following:

$$A \xrightarrow{f+g} B \;=\; A \xrightarrow{\binom{1_A}{1_A}} A \oplus A \xrightarrow{(f\ g)} B \tag{6}$$

Apply the functor, we get:

$$F(A) \xrightarrow{F(f+g)} F(B) \;=\; F(A) \xrightarrow{F\binom{1_A}{1_A}} F(A \oplus A) \xrightarrow{F(f\ g)} F(B) \tag{7}$$

$$= F(A) \xrightarrow{\binom{1_{F(A)}}{1_{F(A)}}} F(A \oplus A) \xrightarrow{(Ff\ Fg)} F(B) \tag{8}$$

$$= F(A) \xrightarrow{Ff+Fg} F(B) \tag{9}$$

Note: Since both projection $F\pi_A : F(A \oplus A) \to F(A)$ satisfy $F\pi_A \circ F\binom{1_A}{1_A} = F\left(\pi_A \circ \binom{1_A}{1_A}\right) = F1_A = 1_{F(A)}$,and $F\pi_A$ are the projections of $F(A \oplus A) \cong F(A) \oplus F(A)$ (as F respects direct sum), then $F\binom{1_A}{1_A} = \binom{1_{F(A)}}{1_{F(A)}}$. Apply similar logic to $F(f\ g)$ we get the same.

So, $F(f + g) = Ff + Fg$, hence F is additive.

$\square$

As a result, we have the following:

Corollary 1. *A left / right exact functor between Abelian Categories is additive.*

Proof. Because the identity relations on inclusion / projections are preserved by any functor, so $F\pi_i \circ F\iota_i = F(\pi_i \circ \iota_i) = F(1_i) = 1_{F(i)}$ given $i = A, B$.

Also, left / right exact functor preserves either kernel or cokernel. For definiteness, if F is left exact, then for $i \neq j$, we have $F\iota_i$ remains as kernel of $F\pi_j$; then, the identity relations guarantee $F\pi_j$ to be epimorphism, hence a cokernel of $F\iota_i$ (which is a kernel). (For right exact functor, it preserves cokernel instead, so swap the proof to $F\pi_j$ works just fine).

By the characterization of direct sum, left / right exact functor preserves direct sum system, hence is additive. $\square$

2 Tools for Embedding

Now, let's define the concept of embedding:

Definition 5. *Let $F : \mathcal{A} \to \mathcal{B}$ be an additive functor. F is **faithful**, if the homomorphism $\mathrm{Hom}_{\mathcal{A}}(A, B) \to \mathrm{Hom}_{\mathcal{B}}(F(A), F(B))$ is injective. In this case, F is called an **embedding**.*

Include some properties of embedding if needed

Here comes some iportant notions for embeddings:

Definition 6. *An object $P \in \mathcal{A}$ is **projective**, if the functor $\mathrm{Hom}_{\mathcal{A}}(P, \_) : \mathcal{A} \to \mathsf{Ab}$ is exact.*

Equivalently, given any epimorphism $f : A \twoheadrightarrow B$, the post composition map $\mathrm{Hom}_{\mathcal{A}}(P, A) \xrightarrow{f\circ\_} \mathrm{Hom}_{\mathcal{A}}(P, B)$ is surjective. It's equivalent to the following:
Given any morphism $p : P \to B$, there exists $p' : P \to A$, such that $p = f \circ p'$, or the following diagram commutes:

$$\begin{tikzcd} P \arrow[d, dashed, "\exists p'"'] \arrow[dr, "p"] & \\ A \arrow[r, two heads, "f"'] & B \end{tikzcd}$$

Which, it has the following property:

Proposition 3. *Let $\{P_i\}_i$ be a family of projective objects in $\mathcal{A}$, if their direct sum $\bigoplus_i P_i$ exists, then it's also projective.*

Proof. Let $f : A \twoheadrightarrow B$ be an epimorphism, and $(p_i)_i : \bigoplus_i P_i \to B$ a morphism. Then, composing with each inclusion, we have $(p_i)_i \circ iota_j = p_j : P_j \to B$ for all index j.

The individual projectiveness guarantees a morphism $p'_j : P_j \to A$ such that $p_j = f \circ p'_j$, hence the direct sum property gathers them into a morphism $(p'_i)_i : \bigoplus_i P_i \to A$, such that $(p'_i)_i \circ \iota_j = p'_j$ for all index j.

Finally, notice that $f \circ (p'_i)_i = (p_i)_i$: Precomposing with all inclusions, we get $f \circ (p'_i)_i \circ \iota_j = f \circ p'_j = p_j = (p_i)_i \circ \iota_j$. Hence, the coproduct propety guarantees the two to be the same. This shows the projectiveness of $\bigoplus_i P_i$. □

Definition 7. *An object $G \in \mathcal{A}$is a **generator**, if the functor $\mathrm{Hom}_{\mathcal{A}}(G, \_) : \mathcal{A} \to \mathsf{Ab}$ is an embedding.*

Here are some characterization of generators:

Proposition 4. *TFAE:*

1. *G is a generator.*
2. *For every nonzero $f : A \to B$, there exists $g : G \to A$, such that $f \circ g : G \to A \to B$ is nonzero.*

3. *Let $\iota : A' \hookrightarrow A$ be a proper subobject, there exists a morphism $g : G \to A$, such that $\operatorname{im}(g)$ is not a subobject of A' (i.e. there doesn't exist a monomorphism $i : \operatorname{im}(g) \hookrightarrow A'$, such that $\operatorname{im}(g) \hookrightarrow A = \operatorname{im}(g) \overset{i}{\hookrightarrow} A' \overset{\iota}{\hookrightarrow} A$).*

Proof.

$1 \iff 2$: Here, G is a generator $\iff$ the map $\operatorname{Hom}_{\mathcal{A}}(A, B) \to \operatorname{Hom}_{\mathsf{Ab}}(\operatorname{Hom}_{\mathcal{A}}(G, A), \operatorname{Hom}_{\mathcal{A}}(G, B))$ by $f \mapsto (f \circ \_)$ is injective for all object A, B.

Hence, given G is a generator, every nonzero $f : A \to B$ has $f \circ \_ : \operatorname{Hom}_{\mathcal{A}}(G, A) \to \operatorname{Hom}_{\mathcal{A}}(G, B)$ being nonzero, there exists $g \in \operatorname{Hom}_{\mathcal{A}}(G, A)$ where $f \circ g \neq 0$.

Conversely, if the second condition is true, then the map $\operatorname{Hom}_{\mathcal{A}}(A, B) \to \operatorname{Hom}_{\mathsf{Ab}}(\operatorname{Hom}_{\mathcal{A}}(G, A), \operatorname{Hom}_{\mathcal{A}}(G, B))$ by $f \mapsto (f \circ \_)$ is injective (as $f \neq 0$ implies $f \circ \_ \neq 0$), hence $\operatorname{Hom}_{\mathcal{A}}(G, \_)$ is an embedding.

$2 \implies 3$: Suppose all nonzero $f : A \to B$ has some $g : G \to A$, such that $f \circ g : G \to A \to B$ is nonzero. Let $\iota : A' \hookrightarrow A$ be a proper subobject, then its cokernel $c : A \twoheadrightarrow C$ is nonzero. Hence, there exists morphism $g : G \to A$, such that $c \circ g : G \to A \twoheadrightarrow C$ is nonzero.

As ι is also a kernel of c (since it's monic), with $c \circ g \neq 0$, g doesn't factor through ι, hence its image is not a subobject of A'.

$2 \impliedby 3$: If condition 3 holds, for any nonzero $f : A \to B$, its kernel $k : K \hookrightarrow A$ is a proper subobject. Hence, there exists $g : G \to A$, such that its image isn't a subobject of K' (equivalently, g doesn't factor through k). As a result, $f \circ g \neq 0$ (as if $f \circ g = 0$, then g must factor through k, which contradicts).

$\square$

Here is one of the key tools for the special case of the embedding theorem - the projective generators:

Lemma 1. *If P is projective, then P is a generator $\iff$ $\operatorname{Hom}_{\mathcal{A}}(P, A)$ is nontrivial for all $A \ncong \mathbf{0}$.*

Proof.

$\implies$: Suppose P is a generator, then for any $A \ncong \mathbf{0}$, because identity $1_A : A \xrightarrow{\sim} A$ is nonzero, then the map $1_A \circ \_ : \operatorname{Hom}_{\mathcal{A}}(P, A) \to \operatorname{Hom}_{\mathcal{A}}(P, A)$ is nonzero. Hence, there exists $g \in \operatorname{Hom}_{\mathcal{A}}(P, A)$ such that $1_A \circ g = g \neq 0$, showing $\operatorname{Hom}_{\mathcal{A}}(P, A) \neq \mathbf{0}$.

$\impliedby$: Suppose $\operatorname{Hom}_{\mathcal{A}}(P, A)$ is nontrivial for all $A \ncong \mathbf{0}$. Here, we use the second condition above for generators: Given nonzero $f : A \to B$, consider the coimage of f, say $c : A \twoheadrightarrow C$ (hence, there exists monomorphism $\overline{f} : C \hookrightarrow B$, such that $f = \overline{f} \circ c$). This forces $c \neq 0$, hence $C \ncong \mathbf{0}$, in particular $\operatorname{Hom}_{\mathcal{A}}(P, C) \neq \mathbf{0}$. So, there exists nontrivial $p : P \to C$.

Now, using the projectiveness of P, there exists $p' : P \to A$, such that $p = c \circ p'$ (which, $p' \neq 0$ because $p \neq 0$). Then, we have $f \circ p' = \overline{f} \circ (c \circ p') = \overline{f} \circ p$. Finally, since $\overline{f}$ is a monomorphism, then $p \neq 0$ implies $\overline{f} \circ p \neq 0$, hence $f \circ p' \neq 0$.

So, the second condition for generator is satisfied, showing P is a generator.

$\square$

With this notion, the category can be more controlled in the following sense:

Definition 8. *A category is **well-powered**, if the family of subobjects (i.e. equivalence classes of monomorphisms $A' \hookrightarrow A$, where the equivalence are stated by isomorphisms on A' that commutes with the monomorphisms) forms a set.*

Proposition 5. *An Abelian Category that has a generator is well-powered.*

Proof. Let $G \in \mathcal{A}$ be a generator, and $A \in \mathcal{A}$ be any object. Then, given a nontrivial subobject $\iota : A' \hookrightarrow A$, the map $\iota \circ \_ : \mathrm{Hom}_{\mathcal{A}}(G, A') \to \mathrm{Hom}_{\mathcal{A}}(G, A)$ is injective.

Hence, given A' a subobject of A, it can be associated to a subgroup of $\mathrm{Hom}_{\mathcal{A}}(G, A)$ (namely the image of $\mathrm{Hom}_{\mathcal{A}}(G, A')$ under $\iota \circ \_$).

This is well-defined, as if $\iota_1 : A'_1 \hookrightarrow A$ and $\iota_2 : A'_2 \hookrightarrow A$ are two isomorphic subobjects, there exists isomorphism $\phi : A'_1 \xrightarrow{\sim} A'_2$ such that $\iota_1 = \iota_2 \circ \phi$. Hence, for any $f \in \mathrm{Hom}_{\mathcal{A}}(G, A'_1)$, $\iota_1 \circ f = \iota_2 \circ (\phi \circ f)$ (with $\phi \circ f \in \mathrm{Hom}_{\mathcal{A}}(G, A'_2)$), and conversely all $g \in \mathrm{Hom}_{\mathcal{A}}(G, A'_2)$ satisfies $\iota_2 \circ g = \iota - 1 \circ (\phi^{-1} \circ g)$ (with$\phi^{-1} \circ g \in \mathrm{Hom}_{\mathcal{A}}(G, A'_1)$). This shows that $\iota_1 \circ \_$ and $\iota_2 \circ \_$ have the same set image, hence isomorphic subobject corresponds to the same subgroup in $\mathrm{Hom}_{\mathcal{A}}(G, A)$.

Finally, any two non-isomorphic subobject can't have the same subgroup in $\mathrm{Hom}_{\mathcal{A}}(G, A)$ $\square$